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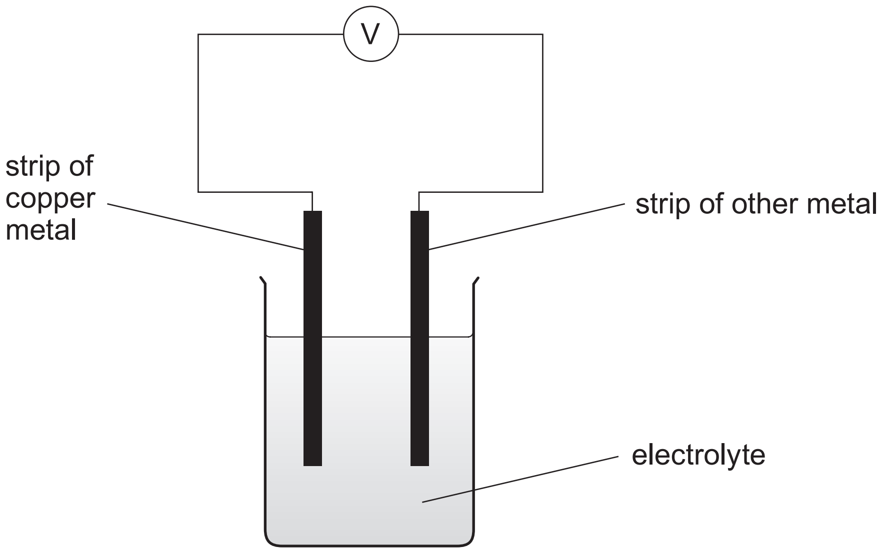
3. When two different metals are connected in a cell, **the metal with the higher reactivity transfers its electrons to the other metal.**

The potential difference produced between pairs of metals can be used to place them in order of reactivity. **The bigger the potential difference, the bigger the difference in reactivity.**

(a) The following apparatus was used to investigate the reactivity of four different metals, **A**, **B**, **C** and **D**, compared with copper.



Each metal was placed separately into a cell with a copper strip. The potential difference was recorded for each metal and the results are shown below.

Metal	Potential difference (V)	Direction of electron flow
A	0.3	copper → metal **A**
B	0.6	metal **B** → copper
C	1.1	metal **C** → copper
D	0.8	copper → metal **D**

Use the information to place metals **A**, **B**, **C** and **D** in order of their reactivity in relation to copper. [2]

most reactive

1. ....
2. ....
3. copper
4. ....
5. ....

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(b) The reactivity of four other metals, **W**, **X**, **Y** and **Z**, was also investigated using the same apparatus. Some of the results are shown in the following table.

Pair of metals in the cell	Potential difference (V)	Direction of electron flow
W and **X**	1.2	**W** → **X**
W and **Y**	0.9	**Y** → **W**
W and **Z**	0.8	
Y and **Z**		**Y** → **Z**

The order of reactivity of these four metals is as follows.

most reactive

Y

Z

W

X

Use this information to give

(i) the direction of the electron flow when **W** and **Z** are placed in the cell, [1]

....

(ii) the potential difference for the cell with metals **Y** and **Z**. [1]

.... V

(c) When copper and zinc are placed into the cell, the following reaction takes place.

$$Cu^{2+} + Zn \longrightarrow Zn^{2+} + Cu$$

Explain how **this** reaction shows both oxidation and reduction. [2]

....

....

....

....

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Turn over.